MIL-A-25719A(USAF)
AMENDMENT 2

9 FEB 84
Superseding
AMENDMENT 1
30 August 1979

MILITARY SPECIFICATION

ACCELEROMETER, MECHANICAL, AIRCRAFT, PILOT'S WARNING, TYPE MA-1

This amendment forms a part of Military Specification MIL-A-25719A(USAF), dated 8 July 1974 and is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

*2.1: Delete entirely and substitute the following:

"2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein."

*2.1: Under "SPECIFICATIONS, Federal", delete "QQ-P-416" and its title. Add the following:

"PPP-B-601 - Boxes, Wood, Cleated Plywood"

*2.1: Under "SPECIFICATIONS, Military", delete "MIL-E-5272" and its title. Add the following:

"MIL-C-83488 - Coating, Aluminum, Ion Vapor Deposited"

Add the following boxed paragraph to bottom of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB Ok 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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*Delete the parenthetical note and add the following:

"2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)"

3.1: Delete and substitute:

"3.1 First article. When specified (see 6.4), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.5)."

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*3.15.2: Delete and substitute:

"3.15.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

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*4.1,b: Delete and substitute:

"b. First article inspection. First article inspections are those tests accomplished on sample units submitted for acceptance as a satisfactory product (see 4.5)."

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*Add the following four paragraphs after 4.3.4:

"4.3.5 Performance record. Prior to conducting the tests described in paragraphs 4.4.6, 4.4.7, 4.4.10, 4.4.13, 4.4.14, 4.4.15, and 4.4.17, the accelerometer shall be subjected to a comprehensive operational test under standard conditions and a record made of all data necessary to determine compliance with this specification. Where the test paragraph establishes the level of acceptable performance, a detailed pre-exposure performance record need not be made. For this latter category, the accelerometer shall be operated prior to test to insure that no malfunction exists.

4.3.6 Installation check. Following installation in the test facility and prior to test, the accelerometer shall be operated sufficiently to insure that no malfunction or damage was caused due to faulty installation procedure or handling.

4.3.7 Criteria for failure. Deterioration or change in performance of any components which could in any manner prevent the accelerometer from meeting functional, maintenance and service requirements during service life shall provide reason to consider the accelerometer as having failed to comply with the conditions of the test to which it was subjected.

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4.3.8 Evaluation of accelerometer. When so directed in individual test procedures, the accelerometer shall be operated to permit performance data to be obtained, or inspected for evidence of deterioration. The performance data under test conditions shall be satisfactorily comparable to that obtained in compliance with 4.3.5. Any deterioration observed shall not exceed that defined in 4.3.7."

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*4.4.6: Delete and substitute:

"4.4.6 Low temperature. The accelerometer shall be placed within the test chamber and the chamber cooled to and maintained at a temperature of -54°C (-65°F) for 4 hours. Upon completion of the test period and while at the test temperature, the accelerometer shall be subjected to and meet the individual tests."

*4.4.7: Delete and substitute:

"4.4.7 High temperature. The accelerometer shall be placed within the test chamber. The internal temperature of the chamber shall then be raised to 71°C (160°F). The accelerometer shall be exposed to that temperature for a period of 4 hours. A relative humidity of not more than 15 percent shall be maintained in the test chamber throughout the exposure period. Upon completion of the test period and while at the test temperature, the accelerometer shall be subjected to and meet the individual tests."

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*4.4.10: Delete and substitute:

"4.4.10 Vibration endurance. The accelerometer shall be mounted on the apparatus in its normal operating position. The accelerometer shall be vibrated with circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly from 5 to 50 Hz and return once each hour for a 50-hour period. Upon completion of the test period, the accelerometer shall be subjeted to the test specified in 4.4.3. The dynamic scale error shall not exceed 0.2g."

*4.4.13: Delete and substitute:

"4.4.13 Low temperature - high altitude. The accelerometer shall be placed within the test chamber and the chamber cooled to and maintained at a temperature of -62°C (-80°F) and at an altitude of 70,000 feet (1.31 inches Hg) for a period of 72 hours, at which time the accelerometer shall be inspected in accordance with the requirements of 4.3.8. The temperature of the chamber shall then be raised to -54°C (-65°F) and maintained for an additional 24-hour period. At the conclusion of this exposure period and while at this temperature, the accelerometer shall be inspected and operated in accordance

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with 4.3.8. The accelerometer temperature shall then be returned to that of standard conditions and the accelerometer operated and inspected visually as specified in 4.3.8. There shall be no damage to the accelerometer as a result of this test."

*4.4.14: Delete and substitute:

"4.4.14 Humidity. The accelerometer shall be placed in the test chamber and set up to simulate installed conditions. The test chamber shall be vented to the atmosphere to prevent the build-up of pressure. Prior to the starting of the test period, the chamber temperature shall be between 20° and 38°C (68° and 100°F) with uncontrolled humidity. During the first 2-hour period, the temperature shall be gradually raised to 71°C (160°F). This temperature shall be maintained during the next 6-hour period. The velocity of the air throughout the test area shall not exceed 150 feet per minute. During the following 16-hour period, the temperature in the chamber shall be gradually reduced to 20° to 38°C (68° to 100°F) which constitutes one cycle. The relative humidity throughout the cycle shall be 95 percent. Steam or distilled water having a pH value between 6.5 and 7.5 at 25°C (77°F) shall be used to obtain the desired humidity. The cycle shall be repeated a sufficient number of times to extend the total time of the test to 240 hours (10 cycles). At the conclusion of the 240 hour period, the accelerometer shall be returned to standard conditions. The accelerometer shall then be operated and inspected within 1 hour as directed in 4.3.8."

*4.4.15: Delete and substitute

"4.4.15 Fungus. Four groups of fungi are listed below and one species of fungus from each group shall be used. In preparing the spore solution, distilled water having a pH value between 5.8 and 7.2 at a temperature between 22° and 32°C (72° and 89°F) shall be utilized. Approximately 10 ml of the solution shall then be introduced directly into a tube culture of the fungus and the spores brought into suspension by vigorous shaking or by gentle rubbing of the spore layer with an innoculating loop without disturbing the agar surface. This process shall be repeated for each species of fungus. The separate spore suspensions from the four types of fungi shall be mixed together to provide a composite suspension. Actively growing cultures between 7 and 21 days old after initial innoculation shall be used for the preparation of the spore suspension. After preparation, the spore suspension will not be kept for more than a 24-hour period at temperatures from 22° to 32°C (72° to 89°F) or not more than 48 hours at temperatures from 2° to 7°C (35° to 45°F). The accelerometer, including applicable external connections, shall be placed in a mold chamber equal to MIL-C-9452, maintaining an external temperature of 30° ± 2° (86° ± 3.6°F) and a relative humidity of 95 ± 5 percent, and sprayed with the suspension of mixed spores. The test period shall be 28 days. At the end of the test, the accelerometer shall be visually examined in accordance with 4.3.8.

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4.4.15.1 Organism.

Group I -	Chaetomium globosum	6205
	Myrothecium verrucaria	9095
Group II -	Memenoniella echinata	9597
	Aspergillus niger	6275
Group III -	Aspergillus flavus	10836
	Aspergillus terreus	10690
Group IV -	Penicillium citrinum	9849
	Penicillium ochrochloron	9112

4.4.15.2 Stock culture designation and source. When ordering cultures, request the cultures by name and serial number. The serial number of the cultures remain the same regardless of source.

Source - American Type Culture Collection
2112 M Street, N.W.,
Washington 6, D.C. "

*4.4.17: Delete and substitute:

"4.4.17 High temperature exposure. The accelerometer shall be placed within the test chamber. The internal temperature of the chamber shall then be raised to 71°C (160°F). The accelerometer shall be exposed to that temperature for a period of 48 hours. A relative humidity of not more than 15 percent shall be maintained in the test chamber throughout the exposure period. At the conclusion of the test period and while still at the test temperature, the accelerometer shall be operated in accordance with 4.3.8. The accelerometer temperature shall then be returned to that of standard conditions and the accelerometer again operated and inspected visually as specified in 4.3.8. No damage to the accelerometer shall result from this test."

*4.5: Delete "Qualification testing" and substitute "First article inspection".

*4.5.1, title and line 1: Delete "qualification" and substitute "First article".

*4.5.2, title, lines 3 and 4: Delete "Qualification" and substitute "First article".

*4.5.3, title and line 1: Delete "Qualification" and substitute "First article".

*4.6: Delete and substitute:

"4.6 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

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5: Delete entire section and substitute:

"5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified by the procuring activity.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

* 5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with method IC-1 of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in 4.6 of this specification.

5.1.2 Level C. Units shall be cleaned, dried and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.3).

* 5.2.1 Level A. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, style A or B, class overseas, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping containers shall be of uniform shape, size and of minimum tare and cube, consistent with the protection required.

* 5.2.2 Level B. Units packaged as specified in 5.1.1 shall be packed in weather-resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

* 5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, each unit and shipping container shall be marked in accordance with MIL-STD-129."

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*6.3: Add the following:

"c. Whether a first article is required for inspection and approval."

*6.4: Delete and substitute:

"6.4 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article."

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*Add new paragraph as follows:

"6.6 International agreement. Certain provisions of this specification are the subject of international standardization agreement ASCC Air Standard 10/4 and STANAG 3330. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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